

Chapter 15 — Capability Architecture

Field	Value
Status	Revised after whole-book constitutional and cross-chapter audit
Version	0.3
Scope	Decouple what HAL intends to accomplish from the replaceable providers and technologies that perform it.
Constitutional basis	Decisions 10, 13–16, 21, 29, 32, 36, 41, 43, 49–50
Owner review	No unresolved Owner decision after whole-book audit

1. Purpose

Decouple what HAL intends to accomplish from the replaceable providers and technologies that perform it.

2. Authoritative Responsibilities

Component	Authoritative responsibility
Capability Catalog	Capability Contract identity, versions, composition and lifecycle
Provider Registry	Current provider identity, availability, health, trust, benchmarks and policy fitness
Adapter Manager	Technical integration, compatibility, sandbox and credential references
Capability Router	Provider choice within policy, privacy, resource, risk and verification constraints

3. Core State and Records

Object	Required content
Capability Contract	Outcome, inputs, outputs, constraints, side effects, quality, permission, risk and evaluation.
Provider Manifest	Claims, versions, dependencies, data use, costs, locality and supported contracts.
Provider Fitness Record	Observed performance, health, trust, calibration, conditions and evidence.
Capability Invocation	Intent, contract version, provider, adapter, execution envelope, transaction and result.

4. Runtime Workflow

1. Owner approves a new capability class through the protected process.
2. Discover provider manifests as claims; authenticate and verify the provider identity.
3. Benchmark and observe capability performance before preferred status.
4. Route an approved request using capability fit, policy, privacy, trust, availability, cost and locality.
5. Invoke through a sandboxed adapter with scoped credentials and transaction context.
6. Evaluate result, update fitness evidence and retire or quarantine unsafe providers.

5. Interfaces and Contracts

All commands, queries and events are typed, versioned, authenticated and correlated. Commands request mutation from the authoritative owner; queries declare consistency and privacy scope; immutable events record completed facts with identity, causation, authorization, provenance and integrity metadata.

6. Failure and Recovery

- Provider outage: route only to compatible verified alternatives or disclose unavailability.
- Adapter compromise: quarantine adapter/provider and revoke scoped credentials.
- Version incompatibility: retain coexistence or block invocation; never guess contract semantics.
- Treaty missing: external provider receives no protected data or invocation.

7. Constitutional Guarantees

- Capabilities extend HAL's reach but never their own authority.
- Provider claims are evidence candidates, not trusted facts.
- Every invocation is attributable to an approved intent, contract, provider and transaction.
- Implementations may change without changing capability semantics or HAL identity.

8. Prohibited Behaviors

- No component may silently broaden authority, reinterpret Owner intent, erase dissent, or treat a derived projection as authoritative state.
- No degraded dependency may silently change policy, identity, trust, authentication, verification or evidence requirements.
- No provider, model, node, Presence, secret, relationship or network location receives constitutional authority by implication.

9. Observability and Verification

The subsystem records correlation, identity, policy/authority context, state versions, decisions, limitations, failures, recovery and outcomes. Verification is risk-scaled and produces durable evidence. Status reporting is derived from governed state rather than model assertion.

10. Source Alignment and Review

This chapter directly implements Decisions 10, 13–16, 21, 29, 32, 36, 41, 43, 49–50. The v0.2 rewrite restores the source-specific objects, workflows and guarantees missing from v0.1. Final disposition is recorded in the separate source-alignment and constitutional-audit reports.

Whole-Book Audit Resolution

The following requirements supersede any ambiguous earlier wording in this chapter and are incorporated into this edition:

- Embedded author approval is not conformance evidence. This chapter's constitutional status is determined only by the independent whole-book audit.
- No substantive constitutional or cross-chapter correction was required in this edition.